

TrES-2b

R-2483

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120723 · created 2026-08-24T00:30:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456131.77538 +/- 0.00017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.021
Transit depth (Rp/Rs)^2	2.22 +/- 0.64 [%]
Orbital Inclination (inc)	85.56 +/- 1.28
Airmass coefficient 1 (a1)	0.033 +/- 0.018
Airmass coefficient 2 (a2)	2.99 +/- 0.12
Scatter in the residuals of the lightcurve fit is	40.95 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.25
Transit Duration (day)	0.093 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.021 vs 0.1254 (deviation 0.2 σ)
Duration fit vs published	0.093 d vs 0.0746 d (deviation 0.26 σ)
Mid-time O–C	0.2 min
Depth SNR	1.3
Residual scatter	40.95 %

Light curve

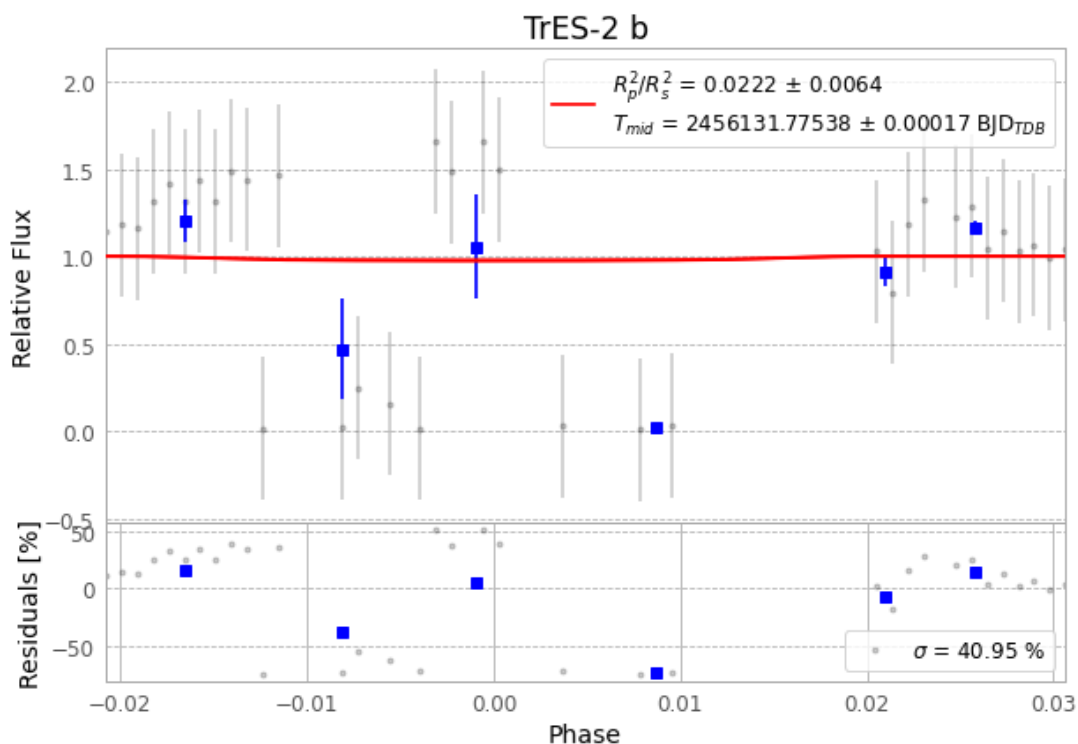


Figure 1 — FinalLightCurve_TrES-2 b_2012-07-22.png (SHA-256 43f2394cbf9ef02b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [102, 229], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ecb727edc1bc698e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), TRES-2120723051812.FITS → TRES-2120723082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120723.log (c5936bae5c64...), FinalLightCurve_TrES-2 b_2012-07-22.png (43f2394cbf9e...), FinalLightCurve_TrES-2 b_2012-07-22.csv (0bafba9c5408...)

Compute resources

Wall time 179.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 00:30Z.